

NAGA VENKATA SAI CHENNU

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Machine Learning Engineer - LLM systems, fine-tuning, and model evaluation

EDUCATION

George Mason University, Fairfax, VA Master of Science in Computer Science GPA 3.57/4.00	Aug 2024 - May 2026
Koneru Lakshmaiah University, Vaddeswaram, India Bachelor of Technology in Computer Science and Engineering First Class with Distinction	Aug 2020 - May 2024

TECHNICAL SKILLS

Production: Python (PyTorch, Hugging Face Transformers, TRL, PEFT, scikit-learn), SQL (PostgreSQL), Docker, Linux, Git
Working: FastAPI, Model Context Protocol (MCP), SLURM, CUDA, CatBoost, TypeScript, AWS, NVIDIA Jetson
Methods: supervised fine-tuning (SFT), LoRA, QLoRA, GRPO, Retrieval-Augmented Generation (RAG), constrained tool calling, eval harness design, error taxonomy, McNemar's exact test, ANOVA

EXPERIENCE

George Mason University, Costello College of Business, Fairfax, VA AI Systems Engineer	Aug 2025 - May 2026
- Traced a TRL v0.20.0 regression that silently dropped assistant-only loss masking by inspecting token-level loss across SFT checkpoints, restoring correct training behavior for every fine-tuning run on the cluster. - Shipped a Claude API extraction service converting 500+ multi-page USPTO and legal filings into JSON by constraining tool-call output to a Pydantic v2 schema rather than repairing malformed output downstream; 95% field-level accuracy against human-labeled ground truth, cutting manual review effort ~80%. - Architected LLM Forge, a configuration-driven 12-stage SFT pipeline supporting LoRA, QLoRA, and GRPO on a 4x NVIDIA H100 SLURM cluster, halving researcher time-to-first-experiment (github.com/Nagavenkatasai7/llm-forge). - Built a 900-check regression suite scored against human-labeled ground truth with McNemar's exact test and ANOVA, surfacing model regressions before release rather than in user reports. - Built the Model Context Protocol (MCP) server routing multi-step tool calls across 2,000+ records, raising extraction throughput ~30%; published the resulting fine-tuned models to Hugging Face for two faculty research groups.	

RESEARCH EXPERIENCE

Koneru Lakshmaiah University, Department of Computer Science and Engineering Research Assistant	Aug 2020 - May 2024
- Benchmarked a four-classifier ensemble against single-model baselines for physiological stress detection, reaching 93% accuracy and 92% F1 with t-test and ANOVA significance testing, not a single held-out split. - Selected CatBoost over an ANN for psychological stress detection on measured accuracy (89%) and precision (91%), then deployed the winning model into a dashboard operated by non-technical reviewers.	

PROJECTS

Vision-Language Chart Understanding - GPT-5.4, ChartX, deterministic eval harness	2026
- Evaluated GPT-5.4 across all 6,000 ChartX images spanning 18 chart types and found accuracy tracked the human perceptual encoding hierarchy: 91-95% on position-encoded charts against 51.6% on treemaps. - Attributed the treemap gap to the benchmark rather than the model: re-rendering treemaps with explicit value labels lifted accuracy 51.6% to 97.7% (McNemar p = 0.0003), showing the ground truth was never recoverable from the image.	
JetBot, On-Device Agent - NVIDIA Jetson Orin Nano, Nemotron, SQLite FTS5, cgroups v2	2026
- Ran a full LLM agent loop on-device under a fixed edge memory budget, isolating inference with cgroups v2 and gating every destructive tool call behind human approval rather than trusting the planner.	

PUBLICATIONS AND PATENT

- Speech Quality Assessment in Indian Languages. IEEE Intl. Conference on Inventive Computation Technologies (ICICT) 2023. doi:10.1109/ICICT57646.2023.10134057 - Three additional first-author papers in IJRITCC (2023) and IJISAE (2023, 2024) on stress detection and AI-driven cybersecurity risk mitigation. - Smart FuelGuard, GPS-integrated ML fuel management. Patent application pending.	
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CERTIFICATIONS

AWS Certified AI Practitioner | NVIDIA-Certified Associate: Generative AI and LLMs | AWS Certified Cloud Practitioner